

International Islamic University Chittagong

Department of Computer Science and Engineering

B.Sc. in CSE, Final Examination, Autumn 2022

Course Code: CSE-2325 Course Title: Digital Logic Design

Time: 2 hours 30 minutes Full Marks: 50

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Group A

[Answer the questions from the followings]

- | | | | | |
|-------|--|-----|---|---|
| 1. a) | Convert SR to T Flip-flop. Convert JK to SR, T to SR, and D to SR Flip-flop. Define the basic difference between Latch & Flip Flop. | CO2 | A | 5 |
| Or, | | | | |
| 1. a) | Design an asynchronous ripple-down counter. | CO2 | A | 5 |
| 1. b) | Design a J-K flip-flop and show its truth table, characteristic table, excitation table, and logic diagram. | CO1 | U | 5 |
| 2. a) | Design a modulo counter by asynchronous counter. | CO2 | U | 5 |
| 2. b) | Design a 5X32 decoder with four 3x8 decoders and a 2x4 decoder. Use a block diagram. | CO3 | A | 5 |
| Or, | | | | |
| 2. b) | Design a circuit that compares two 3-bit numbers, A and B, to check, if they are equal. The circuit has one output x so that x=1 if A=B, and x=0 if A ≠ B. Show the output by providing data into the circuit. | CO3 | A | 5 |

Group B

[Answer the questions from the followings]

- | | | | | |
|-------|--|-----|---|---|
| 3. a) | Design a 2 bits synchronous counter by JK Flip Flop. | CO2 | A | 5 |
| 3. b) | Implement the output Sum of a Half adder by S-R flip flop. | CO3 | N | 5 |
| 4. a) | Design a counter using SR flip flops with the repeated following binary sequence: 0, 1, 3, 2, 6, 4, 5, 7. | CO2 | A | 5 |
| 4. b) | Design Johnson's counter. | CO3 | N | 5 |
| 5. a) | Define ROM. | CO1 | U | 2 |
| 5. b) | Define Moore's state machine with an example. | CO2 | A | 8 |
| Or, | | | | |
| 5. a) | Define register. Define the functions of the universal shift registrar. | CO1 | U | 2 |
| 5. b) | A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers. How many selection inputs are there in each multiplexer? What sizes of multiplexers are needed? How many multiplexers are there on the bus? | CO2 | E | 8 |

International Islamic University Chittagong

Department of Computer Science and Engineering

Final Examination, Autumn 2023

Course Code: CSE 2323

Course Title: Digital Logic Design

Total Marks: 50

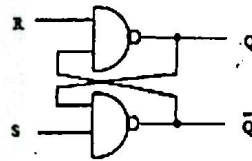
Time: 2 hours 30 minutes

[The figures in the right-hand margin indicate full marks. Course Outcomes and Bloom's Levels are mentioned in additional columns.]

Group-A

[Answer the questions from the followings. Parts of the same question must be answered serially.]

1. a) The figure below shows an RS latch made out of NAND gates (rather than NOR gates). How do Q and Q' (Prime) depend on the RS inputs? i.e. verify that the circuit can indeed be used as a RS latch. CO2 A 5



Or,

- 1 a) Design a SR Flip Flop using NAND gate. CO2 A 5
1. b) Write down the steps associated for SR to D Flip Flop conversion with proper circuit diagram. CO1 U 5
2. a) What do you mean by race condition? How to differentiate between flip-flop and latch? CO2 U 5
2. b) How does JK flip-flop remove the indeterminate states of S-R flip-flop? Design a JK flip-flop and show its characteristic equation, characteristic table, logic diagram and timing diagram. CO3 A 5
- Or
2. b) Describe the operation of master-slave D flip-flop with logic diagram and characteristic table. CO3 A 5

Group-B

3. a) Design a counter with the following binary sequence: 0, 4, 2, 1, 6 and repeat. Use D flip-flops. CO2 A 5
3. b) Design BCD counter with proper timing diagram. What do you mean by shift registers? Mention the difference types of shift register. CO3 N 5
4. a) Design the sequential circuit described by the following state equations using JK flip-flops. CO2 A 5
- $$A(t+1) = xAB + yA'C + xy$$
- $$B(t+1) = xAC + y'BC'$$
- $$C(t+1) = x'B + yAB'$$
4. b) Differentiate between sequential circuit and combinational circuit. CO3 A 5

5. a) Using sequence generator generates sequence of 11010 and also draw proper diagram.

U 5

Or,

5. a) State the characteristics of register. To hold data for longer duration which is used? Define the types of load in register. What are the two types of data used in register?

U 5

5. b) If counter is having 10FF. It is initially 0. What count will it hold after 2060 p
Design D latch with truth table and proper circuit diagram.

A 5

Or,

5. b) Define coincidence logic with example. Differentiate between PAL and PLA
combinational circuit is defined by the functions:

A 5

$$F1(A, B, C) = \sum(3, 5, 6, 7)$$

$$F2(A, B, C) = \sum(0, 2, 4, 7)$$

Implement the circuit with a PLA having three inputs, four product terms, and outputs.

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Semester Final Examination, Spring 2022

Course Code: CSE 2323

Course Title: Digital Logic Design

Time: 2 hours 30 minutes

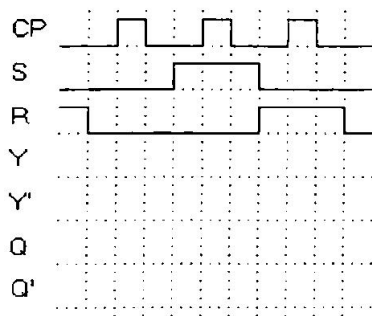
Full Marks: 50

The figures in the right-hand margin indicate full marks

Part A

[Answer the questions from the followings]

- | | | | | | |
|-----|----|---|-----|---|---|
| 1. | a) | Design a look-ahead carry generator for a 3-bit full-adder. | CO2 | A | 5 |
| Or, | | | | | |
| 1. | a) | Design a asynchronous ripple down counter. | CO2 | A | 5 |
| 1. | b) | Design a J-K flip-flop and show its characteristic equation, characteristic table, and logic diagram. | CO1 | U | 5 |
| 2. | a) | Enter the expected timing diagram for the signals Y, Y', Q, and Q' for a master slave S-R flip flop. | CO2 | U | 5 |



- | | | | | | |
|-----|----|--|-----|---|---|
| 2. | b) | Design a 5X32 decoder with four 3x8 decoder and a 2x4 decoder. Use a block diagram. | CO3 | A | 5 |
| Or, | | | | | |
| 2. | b) | Design a circuit that compares two 3 bit numbers, A and B, to check, if they are equal. The circuit has one output x, so that x=1 if A=B, and x=0 if A≠ B. Show the output by providing data into the circuit. | CO3 | A | 5 |

Part B

[Answer the questions from the followings]

- | | | | | | |
|-----|----|---|-----|---|---|
| 3. | a) | Design a sequential circuit with JK flip-flops to satisfy the following state equations:
$A(t+1)=A'B'CD+A'B'C+ACD+AC'D'$
$B(t+1)=A'C+CD'+A'BC'$
$C(t+1)=B$
$D(t+1)=D'$ | CO2 | A | 5 |
| 3. | b) | Design a 2 bits synchronous counter by JK Flip Flop. | CO2 | A | 5 |
| 4. | a) | Design a counter using SR flip-flops with the repeated following binary sequence: 0, 1, 3, 2, 6, 4, 5, 7. | CO2 | A | 5 |
| 4. | b) | Design Johnson's counter | CO3 | N | 5 |
| 5. | a) | Define ROM. | CO1 | U | 2 |
| 5. | b) | Implement the functions $F(w,x,y,z)=\sum(0,1,3,4,8,9,15)$ with ROM. | CO2 | A | 8 |
| Or, | | | | | |
| 5. | a) | Define register. Define the functions of universal shift registrar. | CO1 | U | 3 |
| 5. | b) | A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers. How many selection inputs are there in each multiplexer? What size of multiplexers are needed? How many multiplexers are there in the bus? | CO2 | E | 7 |

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Examination-Spring 2023

Course Code: CSE 2223

Course Title: Digital Logic Design

Time: 2 hours 30 minutes

Full Marks: 50

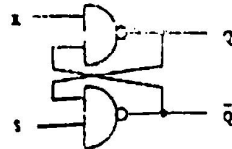
(i) The figures in the right-hand margin indicate full marks

[Answer the questions from the followings. Parts of the same question must be answered serially]

Part A

CO DL Marks

1. a) The figure below shows an RS latch made out of NAND gates (rather than NOR gates). How do Q and Q' (Prime) depend on the RS inputs? i.e. verify that the circuit can indeed be used as a RS latch. CO2 A 5



Or, Design a SR Flip Flop using NAND gate.

1. b) What is flip-flop triggering problem? Describe the master-slave D flip. CO1 U 5
2. a) Differentiate between combinational logic and sequential logic. List some applications of sequential logic. What are the major differences between latch and flip? CO1 U 5
2. b) Design a bus system that would take 4-bits input from two input devices namely keyboard and mouse using multiplexer. CO3 A 5
- Or, How does JK flip-flop remove the indeterminate states of S-R flip-flop? Design a JK flip-flop and show its characteristic equation, characteristic table, logic diagram and timing diagram.

Part B

1. a) Design the sequential circuit described by the following state equations using JK flip-flops. CO2 A 5
- $$A(t+1) = xAB + yA'C + xy$$
- $$B(t+1) = xAC + y'BC'$$
- $$C(t+1) = x'B + yAB'$$
3. b) Design BCD counter with proper timing diagram. CO3 A 5
4. a) Design a 4-bit Up counter with T flip-flop. CO2 A 5
4. b) What are the basic difference between ring counter and Johnson counter? Explain with Block diagram. CO3 R 5
5. a) Using sequence generator generates sequence of 11010 and also draw proper circuit diagram. CO2 A 5
- Or, Design a multiplexer using the following Boolean Equation.
- $$F(A, B, C) = \sum (1, 3, 4, 5, 6, 7)$$
5. b) If counter is having 10FF. It is initially 0. What count will it hold after 2060 pulses? Design D latch with truth table and proper circuit diagram. CO2 A 5
- Or, What is a memory cell? Design a 4x4 RAM and describe its operation with example.

International Islamic University Chittagong

Department of Computer Science and Engineering

Final Examination, Spring 2024

Course Code: CSE 2323

Course Title: Digital Logic Design

Total Marks: 50

Time: 2 hours 30 minutes

- i) The figures in the right-hand margin indicate full marks.
 ii) Course Outcomes and Bloom's Levels are mentioned in additional columns.

Group-A

- a) Construct a 4-to-16 decoder by joining 2-to-4 decoders with enable input. CO2 A 5
 Or
 a) Design a multiplexer using the following Boolean Equation. CO2 A 5

$$F(A, B, C) = \sum (1, 3, 4, 5, 6, 7)$$

 b) What is the carry propagation problem of *parallel* Adder? What could be the solution of carry propagation problem of *parallel* adder? CO1 U 5
 2. a) Explain the working principle of a JK flip-flop and how it differs from an SR flip-flop. Provide a detailed description of the characteristic table and the excitation table for the JK flip-flop. What is the basic building block of sequential logic circuits? CO1 U 5
 b) Write down the differences between latch and flip flop. Write down the steps associated for SR to JK Flip Flop conversion with proper circuit diagram. CO1 U 5

Group-B

3. a) Design a ring counter with proper timing diagram. CO3 A 5
 Or
 Which combinational circuit is renowned for selecting a single input from multiple inputs & directing the binary information to an output line? What is the major functioning responsibility of the multiplexing combinational circuit? 5
 b) Can you explain how to design an 8-to-1 multiplexer using smaller ones? How many inputs does a 4-to-1 multiplexer comprise of? CO2 A 5
 4. a) Design the sequential circuit described by the following state equations using JK flip-flops. CO3 A 5

$$A(t+1) = xAB + yA'C + xy$$

$$B(t+1) = xAC + y'BC'$$

$$C(t+1) = x'B + yAB'$$

 b) Differentiate between sequential circuit and combinational circuit. CO3 10
 i. a) Define coincidence logic with example. Differentiate between PAL and PLA. A
 combinational circuit is defined by the functions:

$$F1(A, B, C) = \sum (3, 5, 6, 7)$$

$$F2(A, B, C) = \sum (0, 2, 4, 7)$$

 - Implement the circuit with a PLA having three inputs, four product terms, and two outputs.
 - What is the main difference between Register and counter?
 Or
 5. a) Design a *universal* shift register and explain its operation. CO3 U 5
 b) Design a 3 bit magnitude comparator circuit with proper explanation. E 5

International Islamic University Chittagong
Department of Computer Science & Engineering

Program: B.Sc.(Engg.) in CSE

Semester: 3rd

Final Examination, Autumn-2023

Course Code: CSE-2324

Course Title: Digital Logic Design Sessional

Time: 3 Hours.

Total Marks: 40

[N. B: Answer any four questions. Each questions carries 10 marks.]

1. To verify the Demorgan's theorems.
2. To verify the Half Adder & Full Adder
3. To verify the characteristic table of JK, and T Flip flops.
4. Study of working principle of 1:4 Demultiplexer using IC-44145
5. To design IC 74193 as a up/down counter
6. Implement BCD to 7- segment Decoder.
7. To design and realize the following using IC 7483. I) BCD to Excess- 3 Code. II) Excess- 3 to BCD Code.
8. To realize One & Two Bit Comparator and study of 7485 magnitude comparator.